

CHARLEVOIX MRC, QC, Canada

WMO: 713190

Lat: 47.284N

Lon: 70.637W

Elev: 2371

StdP: 13.48

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.9	-14.2	-27.1	1.3	-18.3	-21.9	1.8	-13.6	18.9	9.4	16.5	10.4	4.6	310	0.659

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.9	76.0	64.3	73.1	62.3	70.5	60.9	67.1	72.5	65.3	70.3	63.5	68.3	8.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.3	102.0	69.8	63.4	95.4	68.0	61.6	89.5	66.4	33.0	72.8	31.6	70.2	30.1	68.3	75.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.3	14.1	12.2	-25.2	81.3	4.1	3.0	-28.2	83.4	-30.6	85.2	-32.9	86.9	-35.9	89.1		
			-25.6	70.6	4.2	2.1	-28.6	72.1	-31.1	73.4	-33.4	74.6	-36.5	76.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	34.0	6.3	8.7	17.6	30.8	44.4	54.0	59.1	57.8	50.7	37.8	25.7	13.4
	DBStd	20.68	11.85	11.43	11.32	8.35	7.64	7.10	5.38	5.65	7.71	8.06	9.33	10.85
	HDD50	6681	1353	1156	1005	576	210	36	2	6	85	389	728	1135
	HDD65	11349	1818	1576	1469	1025	639	337	195	233	434	845	1178	1600
	CDD50	842	0	0	1	1	37	156	285	248	105	9	0	0
	CDD65	35	0	0	0	0	1	7	13	10	4	0	0	0
	CDH74	183	0	0	0	0	10	49	71	37	16	0	0	0
	CDH80	10	0	0	0	0	0	3	5	1	1	0	0	0
Wind	WSAvg	5.8	5.5	5.9	6.5	6.5	5.9	5.6	5.3	5.1	5.3	6.0	5.9	5.8
Precipitation	PrecAvg	47.8	3.5	3.0	3.2	3.8	4.1	4.3	4.8	4.4	4.3	4.3	4.1	3.9
	PrecMax	58.5	6.9	6.4	5.3	8.8	7.1	7.8	9.8	8.0	8.4	8.2	7.1	7.8
	PrecMin	38.2	1.2	1.4	1.1	1.2	1.5	1.6	0.7	1.8	2.2	1.3	1.0	1.6
	PrecStd	4.7	1.5	1.1	1.1	1.9	1.2	1.5	1.9	1.5	1.3	1.6	1.3	1.5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	41.5	38.8	50.8	59.6	75.2	79.0	79.8	77.9	76.0	64.0	51.9	41.6
		MCWB	40.9	35.3	43.6	48.6	60.3	65.3	66.7	65.8	65.4	53.9	46.8	41.0
	2%	DB	34.2	33.5	41.7	53.1	68.8	74.7	75.6	74.3	71.0	58.4	46.4	35.2
		MCWB	33.5	31.5	36.6	43.3	54.1	62.2	65.1	64.0	62.5	52.5	42.6	34.4
	5%	DB	28.7	29.6	36.8	48.2	64.2	70.9	72.6	71.1	67.4	54.4	42.6	31.8
		MCWB	27.9	27.6	32.5	40.4	52.5	59.7	62.7	61.8	60.9	49.5	39.8	30.9
	10%	DB	23.6	26.3	33.3	43.6	59.4	67.4	69.9	68.5	63.7	50.3	38.7	28.4
		MCWB	22.5	24.7	30.3	37.1	50.3	58.0	61.4	60.8	58.9	46.3	36.3	27.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.9	37.1	47.0	51.0	63.0	68.2	70.3	68.4	67.1	57.3	49.7	40.9
		MCDB	40.3	37.6	50.2	56.6	70.5	75.7	76.4	74.1	72.8	60.6	50.4	41.5
	2%	WB	32.9	31.3	37.2	46.1	58.4	65.0	67.6	66.0	64.7	53.2	44.0	34.7
		MCDB	33.4	32.8	40.5	50.4	65.1	70.5	72.4	71.3	69.4	57.1	45.4	35.0
	5%	WB	27.3	27.8	33.3	41.1	54.8	62.3	65.6	64.3	62.2	50.1	39.8	31.0
		MCDB	28.1	29.3	36.0	47.3	60.2	67.7	70.4	69.0	65.7	53.6	41.4	31.5
	10%	WB	22.3	24.6	30.6	37.6	51.3	59.9	63.4	62.5	59.4	46.7	36.3	27.4
		MCDB	23.2	26.1	33.4	42.7	58.0	65.4	68.4	66.8	63.0	49.5	38.1	28.3
Mean Daily Temperature Range	5% DB	MDBR	17.3	17.9	17.2	15.4	18.3	18.0	16.9	17.0	16.6	13.2	12.1	14.5
		MCDBR	18.8	18.1	18.8	22.0	27.4	23.3	20.3	19.8	19.7	19.4	16.1	15.8
		MCWBR	18.6	16.2	14.8	14.7	15.7	13.3	11.8	11.9	12.8	14.2	13.5	16.0
	5% WB	MCDBR	18.5	16.7	16.7	19.8	21.7	18.6	17.1	16.9	17.2	17.6	14.4	15.5
		MCWBR	18.6	15.8	14.4	14.8	15.1	12.6	11.4	11.2	13.0	14.1	13.6	15.8
Clear-Sky Solar Irradiance	taub		0.237	0.254	0.285	0.317	0.325	0.348	0.364	0.346	0.328	0.303	0.276	0.253
	taud		2.386	2.323	2.311	2.380	2.457	2.419	2.387	2.447	2.492	2.559	2.502	2.390
	Ebn at Noon		278	294	299	298	298	290	283	285	281	274	260	255
	Edh at Noon		21	28	34	35	34	36	36	33	29	23	19	19
All-Sky Solar Radiation	RadAvg		437	726	1097	1409	1605	1728	1676	1459	1138	669	383	297
	RadStd		57	88	88	117	168	115	174	95	91	56	43	28

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (33 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page